PRANEETH KAMSALI

BIM & Digital Construction Portfolio

Revit Modeling | ISO 19650 Compliance | Smart Building Systems

MSc Construction Project Management with BIM
Northumbria University

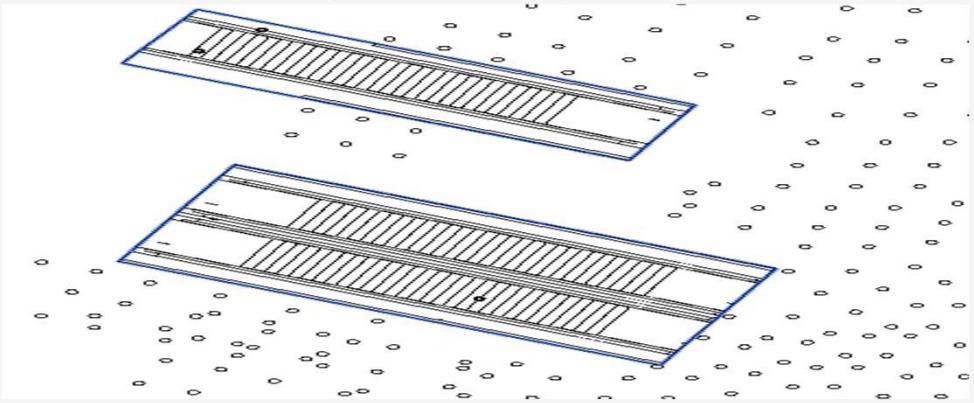
praneeth.kamsali@northumbria.ac.uk
linkedin.com/in/praneethkamsali

BIM Model Improvements: Northumberland Street Store

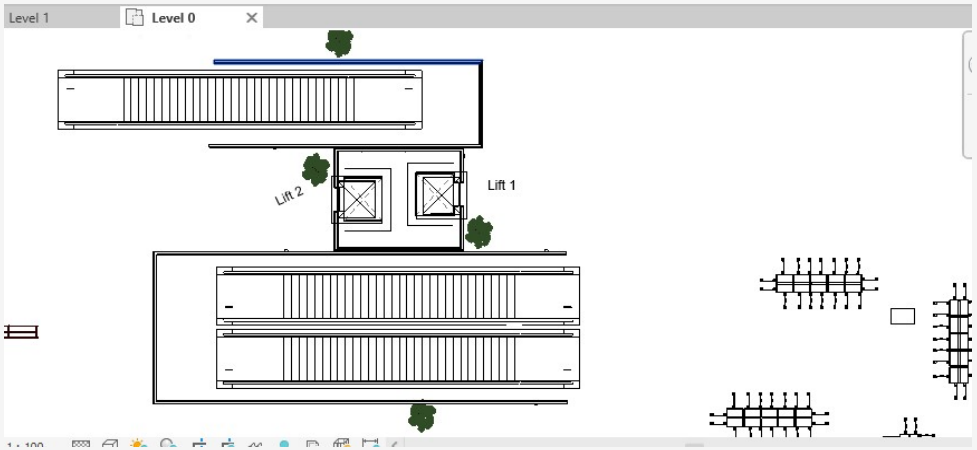
Project: Retail Store Renovation | M&S Northumberland Street

IMPROVEMENT 1: RECEPTION AREA REDESIGN

BEFORE: Original Staircase Layout

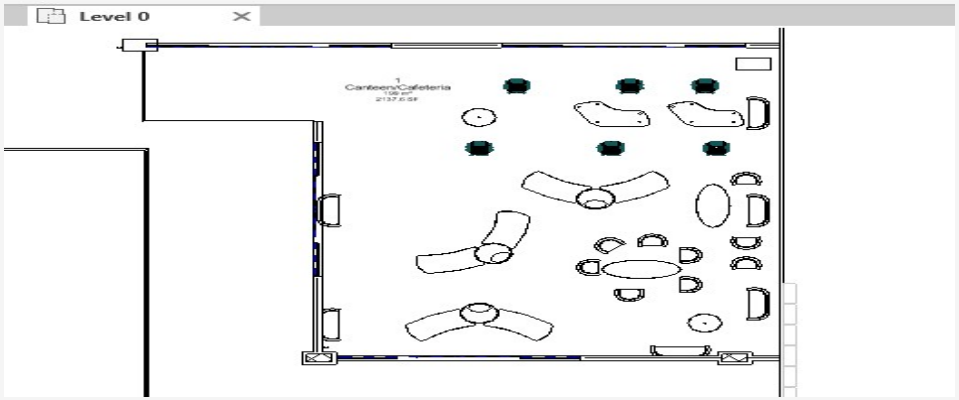


AFTER: Escalators, Lift & Improved Circulation

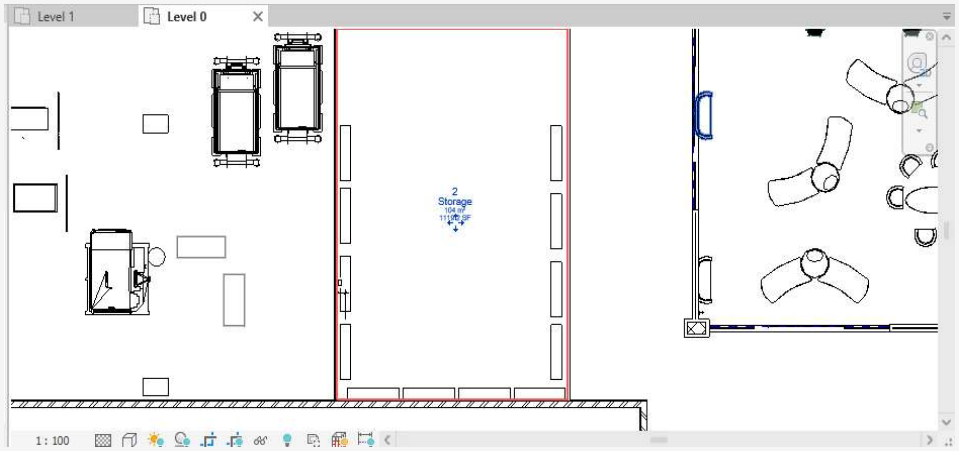


IMPROVEMENT 2: NEW FACILITY ADDITIONS

NEW: Customer Lounge & Resting Areas



NEW: Staff Storage & Service Facilities



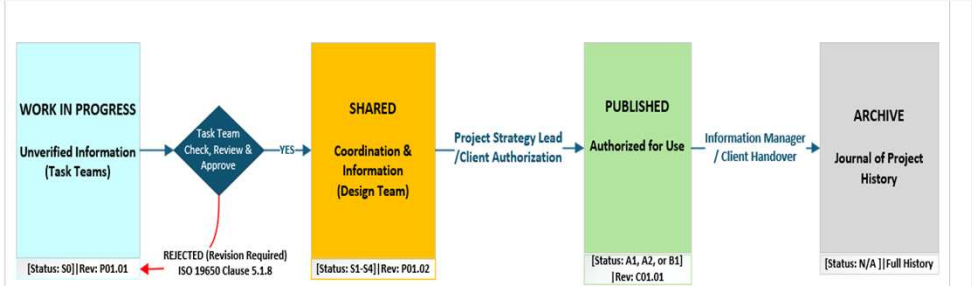
Software: Autodesk Revit | Standards: ISO 19650 | Enhanced accessibility, circulation, and user experience

Design Coordination: Maggie's Centre Extension

Client: Sir Robert McAlpine | £150k Healthcare Facility at Freeman Hospital, Newcastle

PROJECT ROLE

• Supported multidisciplinary BIM model coordination | Coordinated architectural, structural, and MEP disciplines | Implemented ISO 19650 information management | Established Common Data Environment protocols | Applied PAS 1192 naming conventions and workflows



Master Information Delivery Plan (MIDP) - Strategic Extract

ID	File Name (Metadata ID)	Description	Format	Responsibility	Target Date (Stage)
1	MC-UNE-XX-00-M3-A-0001	**Architectural Model (Floating Prisms)**	.RVT	Lead Designer	**Week 4 (S1 Shared)**
2	MC-UNE-XX-00-M3-S-0001	**Structural Model (Screw Piles & Deck)**	.IFC	Structural Eng.	**Week 6 (S2 Shared)**
3	MC-UNE-XX-XX-DR-A-1001	**General Arrangement Plans**	.PDF	Lead Designer	**Week 8 (A1 Published)**
4	MC-UNE-XX-XX-SH-C-2001	**Cost Tracker / EVM Report**	.XLSX	Comm. Manager	**Monthly (S3 Shared)**
5	MC-UNE-XX-XX-VS-A-3001	**Visualization Walkthrough**	.MP4	Vis. Lead	**Week 12 (A1 Published)**

ISO 19650-2 Information State & Suitability Summary

CDE State	Primary Purpose	Suitability Code	Revision Level
WIP	Unverified work within a task team	S0	P01.01
SHARED	Coordination between disciplines	S1-S4	P01.02
PUBLISHED	Authorized construction/tender info	A (or B)	C01.01
ARCHIVE	Legal record of project History	N/A	Full History

ISO 19650 METADATA CONVENTION:
Project-Originator-Volume-Level-Type-Role-Number
(MC-UNE-XX-00-DR-A-0001)

PRANEETH KAMSALI

Mentorship: Aislinn Chambers-Hey (Assistant Design Manager, SRM) | Kevin Alden (Project Manager, SRM)

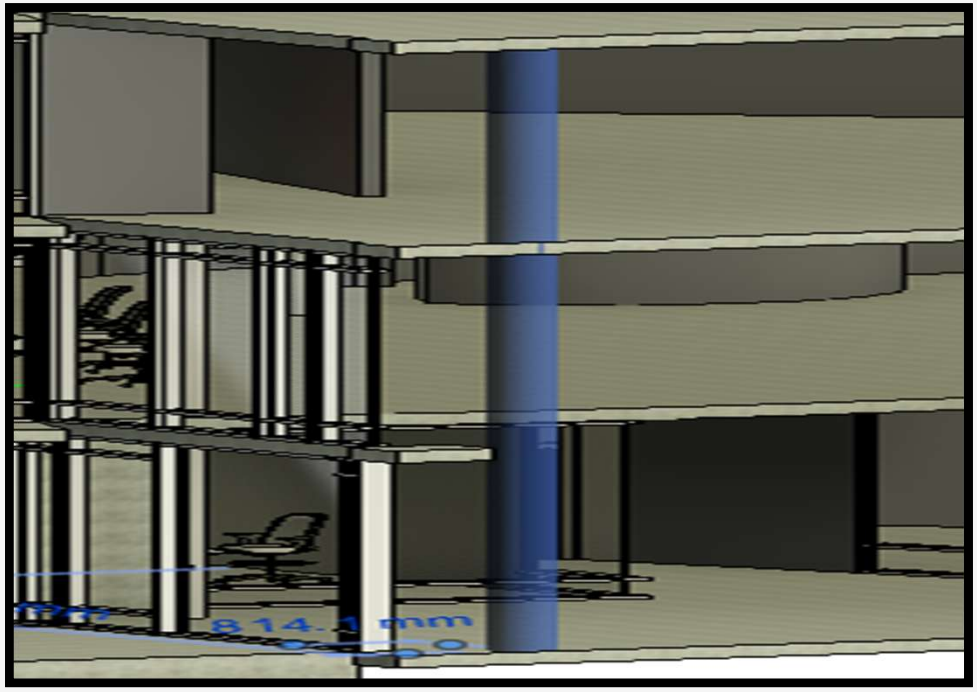
Information Requirements: Building Elements

Detailed BIM Modeling with IFC Schema Compliance | CIS Building Project

ELEMENT 1: ROUNDED COLUMN (Ø450mm)

Specifications:

- Diameter: 450mm | Material: Concrete C30/37
- Fire Resistance: REI90 | GlobalId assigned
- IFC Classification: IfcColumn
- Load-bearing capacity and reinforcement detailed
- Environmental: Embodied carbon data (95.2 kgCO2e)



ELEMENT 2: DOOR (762mm × 2032mm)

Specifications:

- Dimensions: W762 × H2032 × T51mm
- Materials: Aluminium frame, timber panel
- IFC Classification: IfcDoor
- Acoustic Rating: 35 dB | U-value: 1.6 W/m²K
- DDA Compliant | Maintenance: Annual cycle

Technical Standards: IFC 4x2 Schema | LOD 300 | Property Sets (Psets) Populated

ISO 19650 Framework: CIS Building Project

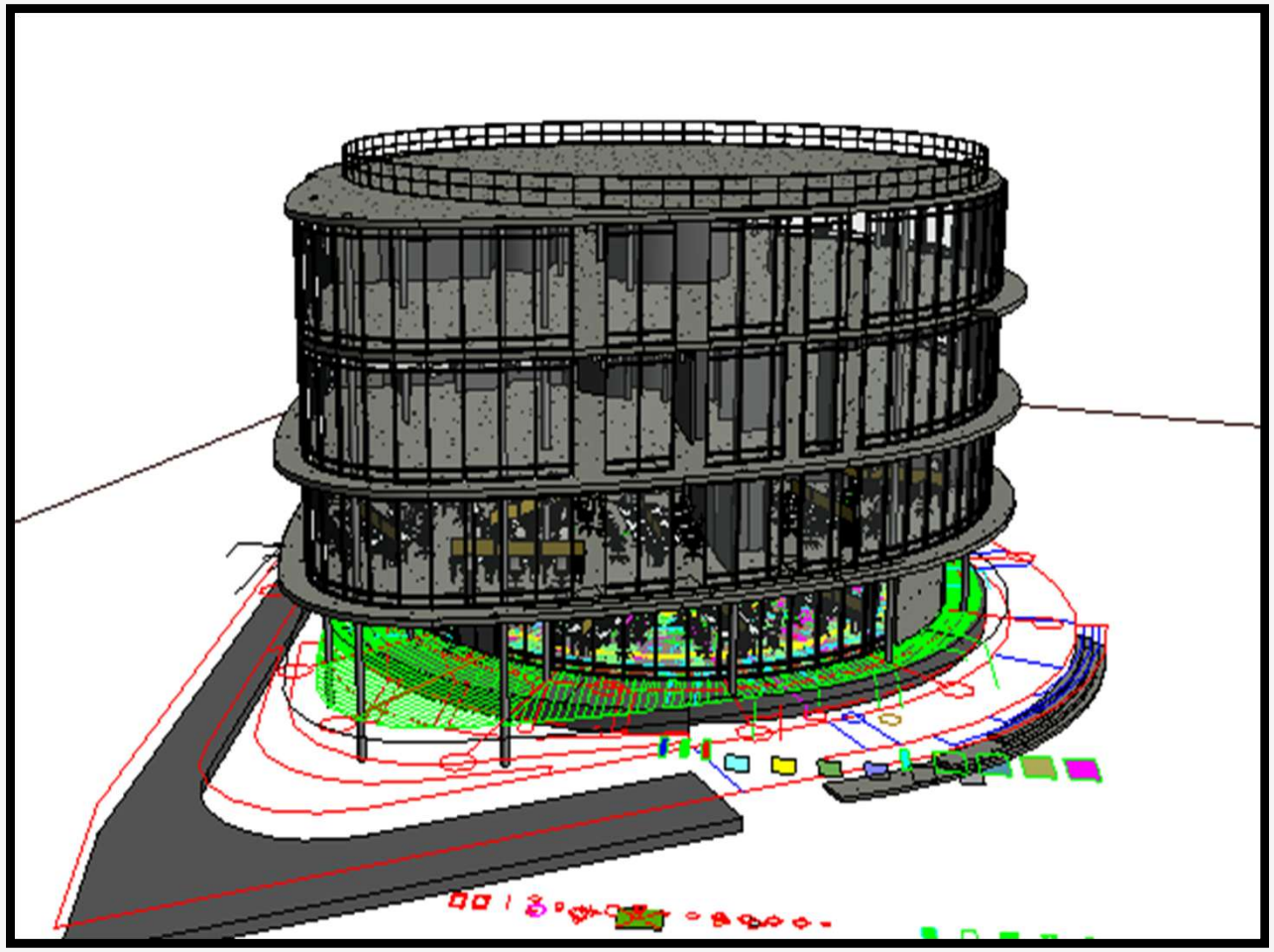
Project: Northumbria University Computer & Information Science Building

COMPLIANCE FRAMEWORK IMPLEMENTATION

- Conceptualized BIM model for ISO 19650-2 compliance from Integrated Project Delivery perspective
- Established Common Data Environment (CDE) protocols with four-state information management
- Implemented structured information workflows aligned with PAS 1192 naming conventions
- Developed Exchange Information Requirements (EIR) and Asset Information Requirements (AIR)
- Applied IFC 4x2 schema for information exchange and data interoperability

TECHNICAL DELIVERABLES

- Information requirements specification for two building elements
- IFC XML data extraction demonstrating schema compliance
- Level of Information Need (LOIN) at LOD 300
- Property sets (Psets) populated for lifecycle management
- Full compliance with ISO 19650-2 standards



Standards: ISO 19650-2 | IFC 4x2 Schema | PAS 1192 Workflows | Academic Research Project

CIS Building Structural View

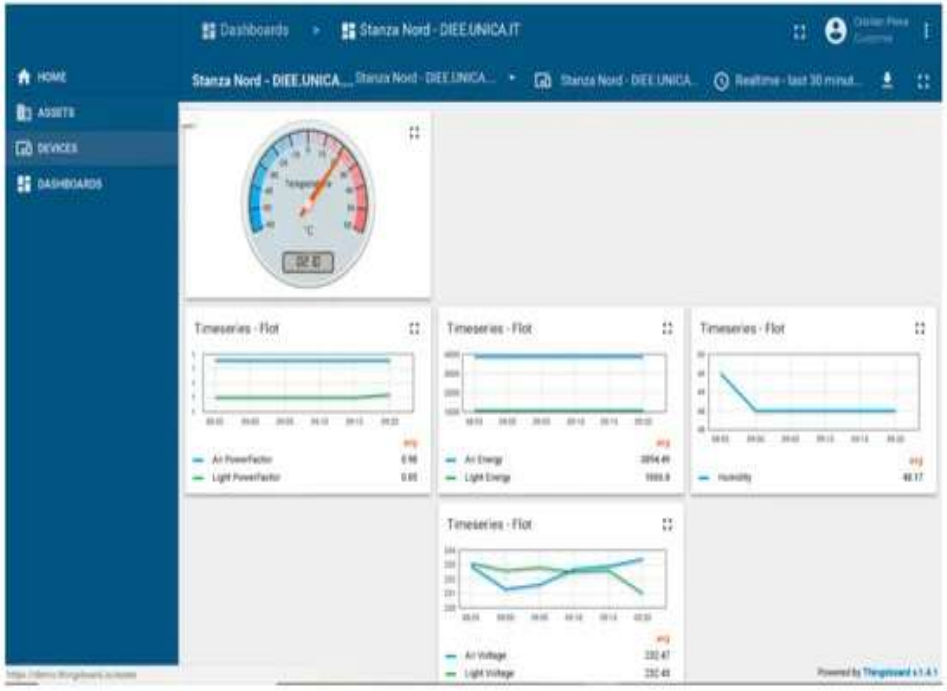
BIM for Facility Management Applications

Operational Efficiency Through Digital Information Management

USE CASE 1: PREDICTIVE MAINTENANCE

- Real-time monitoring with IoT sensors
- Fault Detection & Diagnostics (FDD) systems
- Automated alerts for equipment servicing
- Vibration/energy monitoring for HVAC, escalators

Benefits: Reduced downtime, cost savings, extended asset life

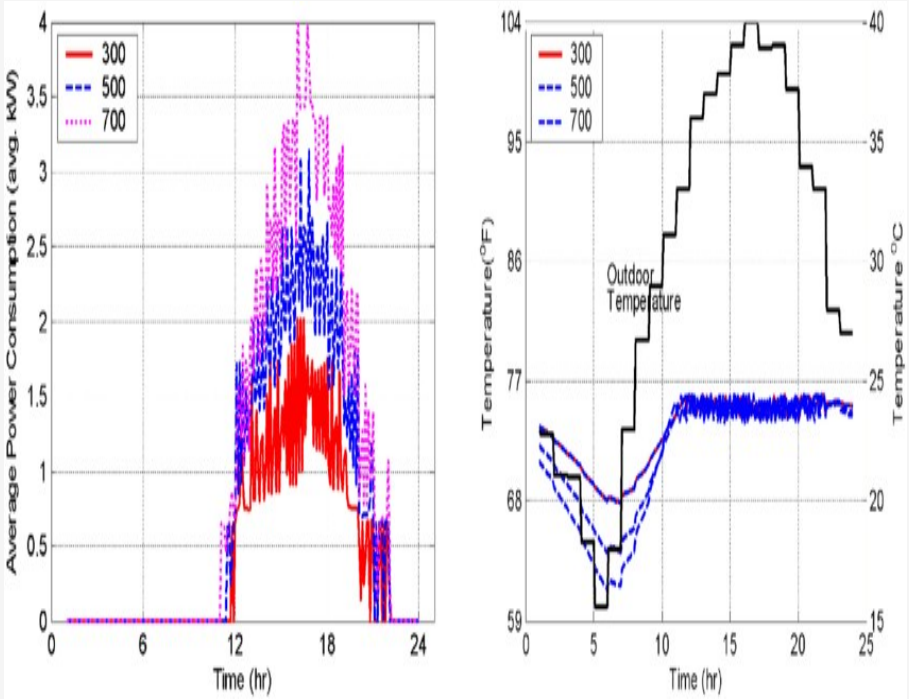


(Source: https://www.mdpi.com)

USE CASE 2: SPACE MANAGEMENT & OPTIMIZATION

- Occupancy analysis with real-time sensors
- Dynamic space allocation and reconfiguration
- Enhanced wayfinding and accessibility
- Data-driven decisions on layout optimization

Benefits: Efficient use, improved flexibility, better UX



(Source: https://www.researchgate.net/)

Platform Integration: BIM + IoT + FM Systems | Real-time Data Analytics

Technical Competencies Demonstrated

SOFTWARE & TOOLS

✓ Autodesk Revit - Model development & coordination
✓ Navisworks - Spatial integration & coordination reviews
✓ AutoCAD - 2D technical drawings
✓ IFC Schema - Information exchange & interoperability
✓ XML - Data structuring & validation

STANDARDS & PROTOCOLS

✓ ISO 19650 - Information management
✓ PAS 1192 - CDE workflows & naming conventions
✓ IFC 4x2 - Industry Foundation Classes schema
✓ Level of Information Need (LOIN) - LOD 300
✓ CDM Regulations - Construction compliance

APPLICATIONS

✓ Digital construction delivery
✓ Multidisciplinary coordination
✓ Facility management integration
✓ Smart building systems
✓ Information management workflows

PROJECTS COMPLETED

- Maggie's Centre Extension (SRM)
- CIS Building (Northumbria University)
- Northumberland Street Store (M&S)
- **Certificate of Recognition Awarded**